

Mars Toktonaliev

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Professional Summary

Senior telecommunications and cloud technology professional with 20+ years of experience in 5G/4G/3G Core networks, cloud architecture, virtualization, and NFV. Proven expertise in designing and deploying carrier-grade telecom solutions in cloud-native environments. CNCF Ambassador with deep knowledge of Kubernetes, cloud orchestration, and telecom networks.

Core Competencies

5G/4G Core Networks: TAS, MSS, IMS, AMF, SMF, UPF, MME and other core network elements.

Cloud & Virtualization: Cloud-native architecture, Kubernetes, OpenStack, VMware, multi-cloud orchestration, ETSI MANO

Network Functions: VNF/CNF deployment, ONAP, network automation, performance benchmarking, troubleshooting

Leadership: Team supervision, cross-functional collaboration, vendor management, technical mentorship

Professional Experience

KGPCo - Irving, TX

Senior System Engineer, Solution Innovation Center

Oct 2019 - Present

Engineering lead for Edge Cloud and telco network functions, driving innovation in cloud-native telecommunications infrastructure:

- Lead VNF deployment, integration, and validation initiatives, including customer demos and Proof of Concepts for next-generation network services
- Architect and implement platform performance benchmarking solutions with comprehensive monitoring infrastructure (Prometheus, Grafana, LibreNMS)
- Optimize integration workflows, reducing whitebox switches pre-production integration time by 25%
- Design and validate edge cloud solutions for carrier-grade telecommunications applications
- Collaborate with cross-functional teams to ensure service SLAs meet business requirements

Nokia - Multiple Locations

5G Cloud RAN System Engineer, Nokia Networks

Irving, TX — Feb 2017 - Oct 2019

Technical lead for cloudRAN initiatives supporting AT&T's 5G Edge Cloud deployment strategy:

- Led ONAP (Open Network Automation Platform) implementation including installation, configuration, troubleshooting, and training programs. Successfully deployed the first complete ONAP solution in Nokia North America and integrated 5G CloudBTS
- Core member of 5G CloudBTS containerization R&D team, successfully establishing the first 5G call on prototype containerized base station software
- Architected cloud-native solutions for 5G network functions ensuring high availability and performance
- Provided technical expertise for cloud orchestration and automation in carrier-grade telecommunications environment
- Collaborated with technology partners and OEMs to deliver best-in-class 5G solutions

Circuit Switch Core Specialist, Nokia Solutions and Networks *Irving, TX — 2011 - Feb 2017*

Key member of ten-person Shared Reference Network team supporting global telecommunications operations:

- Served as subject-matter expert on virtualization, VMware, and OpenStack for core network infrastructure. Increased virtual network elements deployment from 28% to 76% over three years, establishing best practices adopted across global lab locations
- Supervised installation, deployment, and troubleshooting of 100+ OpenStack cloud environments for telecommunications workloads
- Managed virtualized ATCA network elements including vMME, vTAS, and vHLR, ensuring optimal performance and availability
- Supported operation, maintenance, upgrades, and troubleshooting of circuit switch core equipment (MSS, MGW, TAS, BGW)
- Led parameter optimization initiatives to improve network performance across core network elements

Technical Support Engineer, Nokia Siemens Networks *Budapest, Hungary — 2008 - 2011*

- Provided Tier 3 technical support for CS core network elements in large-scale carrier networks
- Validated software patches and feature deployments ensuring quality standards before customer release
- Performed root cause analysis and troubleshooting for complex network issues

Circuit Switch Core Engineer, Nokia Siemens Networks *Bishkek, Kyrgyzstan — 2007 - 2008*

- Delivered Tier 1 technical support for CS core network elements
- Executed network installations, commissioning, integrations, and upgrades for telecommunications operators

Open Source & Community Leadership

- **CNCF Ambassador** since 2017 - Representing cloud-native technologies in telecommunications sector
- **Kubernetes Contributor** - SIG-contribex contributor and mentor, Kubernetes Contributor Summit co-organizer
- **Certification Developer** - Linux Foundation/CNCF KCNA, CKAD, and PCA exams co-author
- **Conference Leadership** - KubeCon track chair (EU21, NA22, NA25) and Kubernetes on Edge conference chair since 2022.
- **OPNFV Contributor** - Virtual switch performance (vsperf) project for NFV optimization
- **Community Organizer** - Founder of Cloud Native Dallas/Dallas Kubernetes meetup groups, lead co-organizer of Kubernetes Community Day Texas (www.texaskcd.com)
- **OpenStack Advocate** - Upstream Institute volunteer, training coordinator for OpenStack Summit participants

Education

MS, Communication Networks and Switching Systems

Kyrgyz National Technical University - Bishkek, Kyrgyzstan

BS, Communication Networks and Switching Systems

Kyrgyz National Technical University - Bishkek, Kyrgyzstan

Technical Certifications & Recognition

CNCF Certified Kubernetes Application Developer (CKAD), Kubernetes Cloud Native Associate (KCNA), Prometheus Certified Associate (PCA) exam developer and subject matter expert